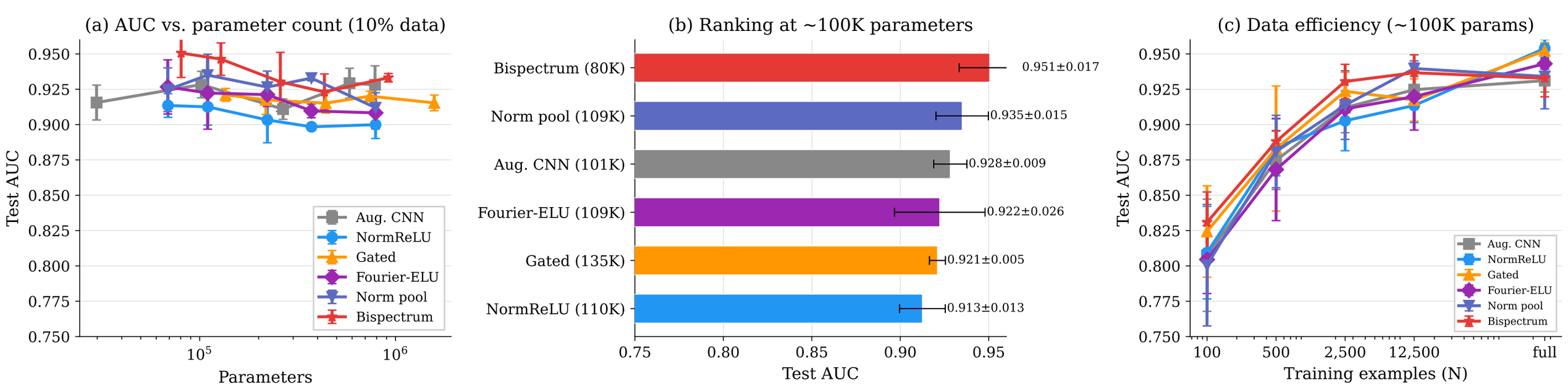
(a) AUC vs. parameter count (10% data)
Test AUC
0.950
0.925
0.900
0.875
0.850
0.825
0.800
0.775
0.750
10^5
10^6
Parameters
Aug. CNN
NormReLU
Gated
Fourier-ELU
Norm pool
Bispectrum
(b) Ranking at ~100K parameters
Bispectrum (80K)
Norm pool (109K)
Aug. CNN (101K)
Fourier-ELU (109K)
Gated (135K)
NormReLU (110K)
0.951±0.017
0.935±0.015
0.928±0.009
0.922±0.026
0.921±0.005
0.913±0.013
0.75
0.80
0.85
0.90
0.95
Test AUC
(c) Data efficiency (~100K params)
Test AUC
0.950
0.925
0.900
0.875
0.850
0.825
0.800
0.775
0.750
100
500
2,500
12,500
full
Training examples (N)
Aug. CNN
NormReLU
Gated
Fourier-ELU
Norm pool
Bispectrum